TP- implement HSRP

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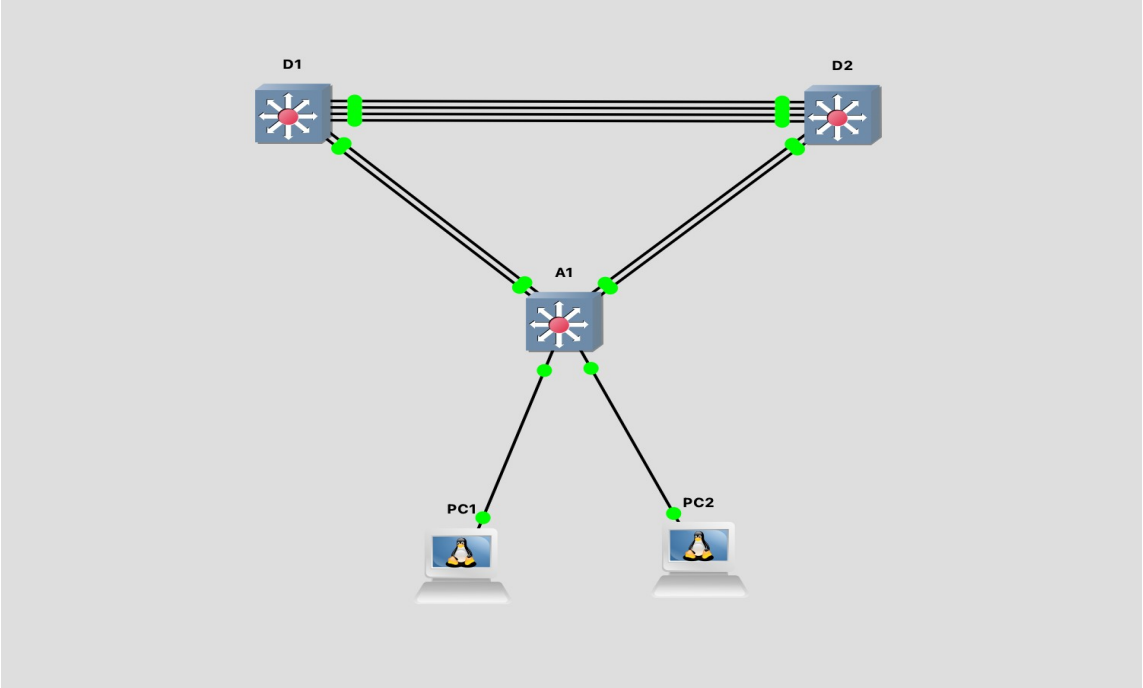
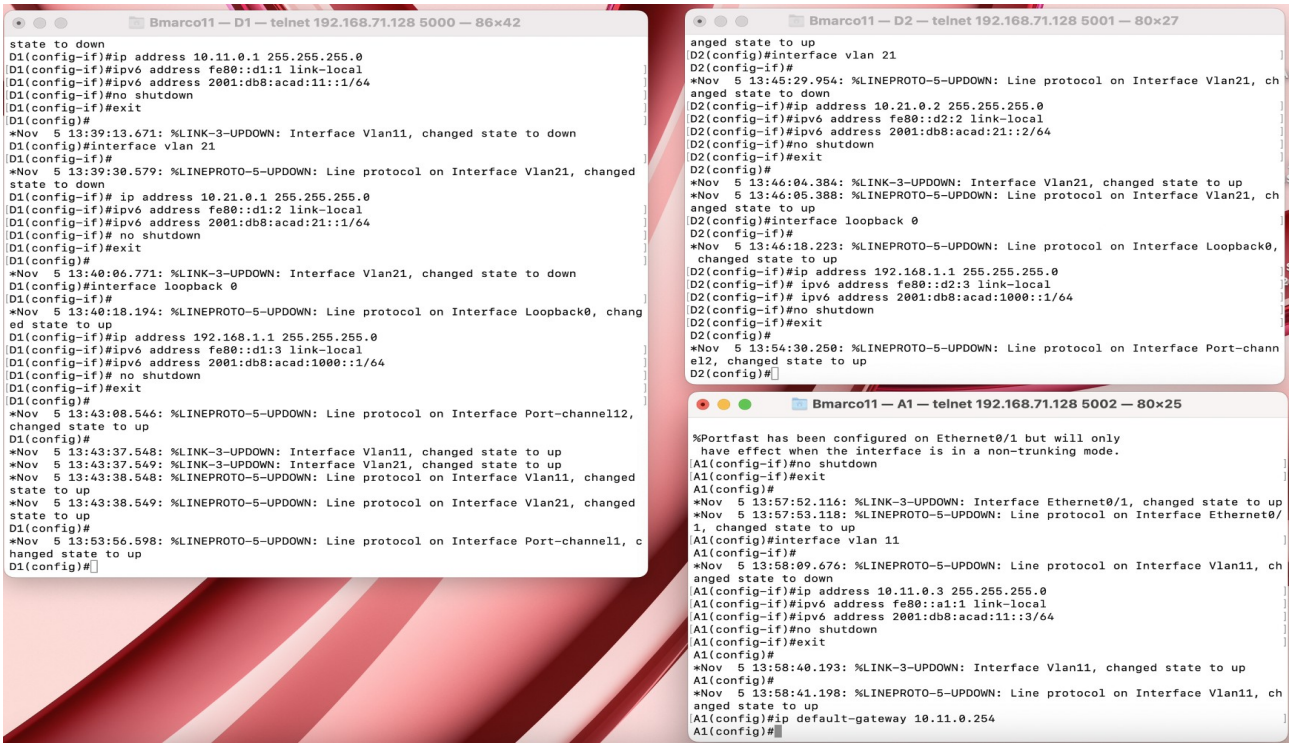
Topology





Bmarco11 — D1 — telnet 192.168.71.128 5000 — 86×42

```
state to down
D1(config-if)#ip address 10.11.0.1 255.255.255.0
D1(config-if)#ipv6 address fe80::d1:1 link-local
D1(config-if)#ipv6 address 2001:db8:acad:11::1/64
D1(config-if)#no shutdown
D1(config-if)#exit
D1(config)#
*Nov  5 13:39:13.671: %LINK-3-UPDOWN: Interface Vlan11, changed state to down
D1(config)#interface vlan 21
D1(config-if)#
*Nov  5 13:39:30.579: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan21, changed
state to down
D1(config-if)# ip address 10.21.0.1 255.255.255.0
D1(config-if)#ipv6 address fe80::d1:2 link-local
D1(config-if)#ipv6 address 2001:db8:acad:21::1/64
D1(config-if)# no shutdown
D1(config-if)#exit
D1(config)#
*Nov  5 13:40:06.771: %LINK-3-UPDOWN: Interface Vlan21, changed state to down
D1(config)#interface loopback 0
D1(config-if)#
*Nov  5 13:40:18.194: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, chang
ed state to up
D1(config-if)#ip address 192.168.1.1 255.255.255.0
D1(config-if)#ipv6 address fe80::d1:3 link-local
D1(config-if)#ipv6 address 2001:db8:acad:1000::1/64
D1(config-if)# no shutdown
D1(config-if)#exit
D1(config)#
*Nov  5 13:43:08.546: %LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel12,
changed state to up
D1(config)#
*Nov  5 13:43:37.548: %LINK-3-UPDOWN: Interface Vlan11, changed state to up
*Nov  5 13:43:37.549: %LINK-3-UPDOWN: Interface Vlan21, changed state to up
*Nov  5 13:43:38.548: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan11, changed
state to up
*Nov  5 13:43:38.549: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan21, changed
state to up
D1(config)#
*Nov  5 13:53:56.598: %LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, c
hanged state to up
D1(config)#
```

Bmarco11 — D2 — telnet 192.168.71.128 5001 — 80×27

```
anged state to up
D2(config)#interface vlan 21
D2(config-if)#
*Nov  5 13:45:29.954: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan21, ch
anged state to down
D2(config-if)#ip address 10.21.0.2 255.255.255.0
D2(config-if)#ipv6 address fe80::d2:2 link-local
D2(config-if)#ipv6 address 2001:db8:acad:21::2/64
D2(config-if)#no shutdown
D2(config-if)#exit
D2(config)#
*Nov  5 13:46:04.384: %LINK-3-UPDOWN: Interface Vlan21, changed state to up
*Nov  5 13:46:05.388: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan21, ch
anged state to up
D2(config)#interface loopback 0
D2(config-if)#
*Nov  5 13:46:18.223: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0,
 changed state to up
D2(config-if)#ip address 192.168.1.1 255.255.255.0
D2(config-if)# ipv6 address fe80::d2:3 link-local
D2(config-if)# ipv6 address 2001:db8:acad:1000::1/64
D2(config-if)#no shutdown
D2(config-if)#exit
D2(config)#
*Nov  5 13:54:30.250: %LINEPROTO-5-UPDOWN: Line protocol on Interface Port-chann
el2, changed state to up
D2(config)#
```

Bmarco11 — A1 — telnet 192.168.71.128 5002 — 80×25

```
%Portfast has been configured on Ethernet0/1 but will only
 have effect when the interface is in a non-trunking mode.
A1(config-if)#no shutdown
A1(config-if)#exit
A1(config)#
*Nov  5 13:57:52.116: %LINK-3-UPDOWN: Interface Ethernet0/1, changed state to up
*Nov  5 13:57:53.118: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
1, changed state to up
A1(config)#interface vlan 11
A1(config-if)#
*Nov  5 13:58:09.676: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan11, ch
anged state to down
A1(config-if)#ip address 10.11.0.3 255.255.255.0
A1(config-if)#ipv6 address fe80::a1:1 link-local
A1(config-if)#ipv6 address 2001:db8:acad:11::3/64
A1(config-if)#no shutdown
A1(config-if)#exit
A1(config)#
*Nov  5 13:58:40.193: %LINK-3-UPDOWN: Interface Vlan11, changed state to up
A1(config)#
*Nov  5 13:58:41.198: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan11, ch
anged state to up
A1(config)#ip default-gateway 10.11.0.254
A1(config)#
```

Bmarco11 — root@PC1: ~ — telnet 192.168.71.128 5003 — 80×24

```
Connected to 192.168.71.128.
Escape character is '^]'.

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.11.0.50  netmask 255.255.255.0  broadcast 0.0.0.0
        inet6 fe80::42:86ff:fe23:a900  prefixlen 64  scopeid 0x20<link>
        inet6 2001:db8:acad:11:42:86ff:fe23:a900  prefixlen 64  scopeid 0x0<glob
al>
        ether 02:42:86:23:a9:00  txqueuelen 1000  (Ethernet)
        RX packets 403  bytes 31728 (31.7 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 7  bytes 602 (602.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

root@PC1:~#
```

Bmarco11 — root@PC2: ~ — telnet 192.168.71.128 5005 — 80×24

```
Escape character is '^]'.
PC2 console is now available... Press RETURN to get started.
root@PC2:~# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.21.0.50  netmask 255.255.255.0  broadcast 0.0.0.0
        inet6 2001:db8:acad:21:42:f1ff:fe0b:7400  prefixlen 64  scopeid 0x0<glob
al>
        inet6 fe80::42:f1ff:fe0b:7400  prefixlen 64  scopeid 0x20<link>
        ether 02:42:f1:0b:74:00  txqueuelen 1000  (Ethernet)
        RX packets 7  bytes 536 (536.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 7  bytes 602 (602.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

root@PC2:~#
```

Bmarco11 — D1 — telnet 192.168.71.128 5000 — 86×22

```
D1#
*Nov  5 14:42:54.825: %HSRP-5-STATECHANGE: Vlan21 Grp 216 state Speak -> Standby
D1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
D1(config)#interface vlan 21
D1(config-if)#standby version 2
D1(config-if)#standby 21 ip 10.21.0.254
D1(config-if)#standby 21 preempt
D1(config-if)#standby 216 ipv6 autoconfig
D1(config-if)#standby 216 preempt
D1(config-if)#end
D1#
*Nov  5 14:46:03.304: %SYS-5-CONFIG_I: Configured from console by console
D1#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State   Active          Standby         Virtual IP
Vl11        11   150 P Active  local           10.11.0.2       10.11.0.254
Vl11        116  150 P Active  local           FE80::D2:1      FE80::5:73FF:FEA0:74
Vl21        21   100 P Standby 10.21.0.2       local           10.21.0.254
Vl21        216  100 P Standby FE80::D2:2      local           FE80::5:73FF:FEA0:D8
D1#
```

Bmarco11 — D2 — telnet 192.168.71.128 5001 — 80×27

```
*Nov  5 14:41:37.608: %HSRP-5-STATECHANGE: Vlan11 Grp 116 state Speak -> Standby
D2(config)#interface vlan 21
D2(config-if)#standby version 2
D2(config-if)#standby 21 ip 10.21.0.254
D2(config-if)#standby 21 priority 150
D2(config-if)#standby 21 preempt
D2(config-if)#
*Nov  5 14:42:17.715: %HSRP-5-STATECHANGE: Vlan21 Grp 21 state Speak -> Active
D2(config-if)#standby 216 ipv6 autoconfig
D2(config-if)#standby 216 priority 150
D2(config-if)#standby 216 preempt
D2(config-if)#
*Nov  5 14:42:44.010: %HSRP-5-STATECHANGE: Vlan21 Grp 216 state Speak -> Active
D2(config-if)#end
D2#
*Nov  5 14:42:54.830: %SYS-5-CONFIG_I: Configured from console by console
D2#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State   Active          Standby         Virtual IP
Vl11        11   100 P Standby 10.11.0.1       local           10.11.0.254
Vl11        116  100 P Standby FE80::D1:1      local           FE80::5:73FF:FEA0
:74
Vl21        21   150 P Active  local           10.21.0.1       10.21.0.254
Vl21        216  150 P Active  local           FE80::D1:2      FE80::5:73FF:FEA0
:D8
D2#
```

Bmarco11 — root@PC1: ~ — telnet 192.168.71.128 5003 — 80×22

```
Last login: Wed Nov  5 15:35:57 on ttys003

64 bytes from 192.168.1.1: icmp_seq=255 ttl=255 time=0.956 ms
64 bytes from 192.168.1.1: icmp_seq=256 ttl=255 time=2.57 ms
64 bytes from 192.168.1.1: icmp_seq=257 ttl=255 time=1.11 ms
64 bytes from 192.168.1.1: icmp_seq=258 ttl=255 time=1.63 ms
64 bytes from 192.168.1.1: icmp_seq=259 ttl=255 time=1.21 ms
64 bytes from 192.168.1.1: icmp_seq=260 ttl=255 time=1.11 ms
64 bytes from 192.168.1.1: icmp_seq=261 ttl=255 time=1.60 ms
64 bytes from 192.168.1.1: icmp_seq=262 ttl=255 time=2.90 ms
64 bytes from 192.168.1.1: icmp_seq=263 ttl=255 time=0.863 ms
64 bytes from 192.168.1.1: icmp_seq=264 ttl=255 time=1.44 ms
64 bytes from 192.168.1.1: icmp_seq=265 ttl=255 time=0.954 ms
64 bytes from 192.168.1.1: icmp_seq=266 ttl=255 time=2.00 ms
64 bytes from 192.168.1.1: icmp_seq=267 ttl=255 time=1.14 ms
64 bytes from 192.168.1.1: icmp_seq=268 ttl=255 time=1.66 ms
64 bytes from 192.168.1.1: icmp_seq=269 ttl=255 time=2.16 ms
64 bytes from 192.168.1.1: icmp_seq=270 ttl=255 time=1.13 ms
64 bytes from 192.168.1.1: icmp_seq=271 ttl=255 time=1.09 ms
^C
--- 192.168.1.1 ping statistics ---
271 packets transmitted, 228 received, 15.8672% packet loss, time 272464ms
rtt min/avg/max/mdev = 0.545/1.518/13.452/1.302 ms
root@PC1:~#
```

Bmarco11 — D1 — telnet 192.168.71.128 5000 — 86×23

```
D1(config-vlan)#exit
D1(config)#interface vlan 11
D1(config-if)#shutdown
D1(config-if)#
*Nov  5 14:50:57.190: %HSRP-5-STATECHANGE: Vlan11 Grp 116 state Active -> Init
D1(config-if)#
*Nov  5 14:50:59.196: %LINK-5-CHANGED: Interface Vlan11, changed state to administrati
vely down
*Nov  5 14:51:00.201: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan11, changed
state to down
*Nov  5 14:51:00.201: %HSRP-5-STATECHANGE: Vlan11 Grp 11 state Active -> Init
D1(config-if)#exit
D1(config)#interface vlan 11
D1(config-if)#no shutdown
D1(config-if)#exit
*Nov  5 14:51:20.290: %LINK-3-UPDOWN: Interface Vlan11, changed state to up
*Nov  5 14:51:21.296: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan11, changed
state to up
D1(config-if)#exit
D1(config)#
*Nov  5 14:51:24.145: %HSRP-5-STATECHANGE: Vlan11 Grp 116 state Listen -> Active
*Nov  5 14:51:24.578: %HSRP-5-STATECHANGE: Vlan11 Grp 11 state Listen -> Active
D1(config)#
```

Bmarco11 — root@PC1: ~ — telnet 192.168.71.128 5003 — 80×24

```
64 bytes from 192.168.1.1: icmp_seq=38 ttl=255 time=3.14 ms
64 bytes from 192.168.1.1: icmp_seq=39 ttl=255 time=1.74 ms
64 bytes from 192.168.1.1: icmp_seq=40 ttl=255 time=1.47 ms
64 bytes from 192.168.1.1: icmp_seq=41 ttl=255 time=1.23 ms
64 bytes from 192.168.1.1: icmp_seq=42 ttl=255 time=1.50 ms
64 bytes from 192.168.1.1: icmp_seq=43 ttl=255 time=1.66 ms
64 bytes from 192.168.1.1: icmp_seq=44 ttl=255 time=3.03 ms
64 bytes from 192.168.1.1: icmp_seq=45 ttl=255 time=2.25 ms
64 bytes from 192.168.1.1: icmp_seq=46 ttl=255 time=1.42 ms
64 bytes from 192.168.1.1: icmp_seq=47 ttl=255 time=2.68 ms
64 bytes from 192.168.1.1: icmp_seq=48 ttl=255 time=2.40 ms
64 bytes from 192.168.1.1: icmp_seq=49 ttl=255 time=1.42 ms
64 bytes from 192.168.1.1: icmp_seq=50 ttl=255 time=2.12 ms
64 bytes from 192.168.1.1: icmp_seq=51 ttl=255 time=1.81 ms
64 bytes from 192.168.1.1: icmp_seq=52 ttl=255 time=2.09 ms
64 bytes from 192.168.1.1: icmp_seq=53 ttl=255 time=1.31 ms
64 bytes from 192.168.1.1: icmp_seq=54 ttl=255 time=2.13 ms
64 bytes from 192.168.1.1: icmp_seq=55 ttl=255 time=7.00 ms
64 bytes from 192.168.1.1: icmp_seq=56 ttl=255 time=1.70 ms
64 bytes from 192.168.1.1: icmp_seq=57 ttl=255 time=5.29 ms
64 bytes from 192.168.1.1: icmp_seq=58 ttl=255 time=1.63 ms
64 bytes from 192.168.1.1: icmp_seq=60 ttl=255 time=1.73 ms
64 bytes from 192.168.1.1: icmp_seq=61 ttl=255 time=0.840 ms
64 bytes from 192.168.1.1: icmp_seq=62 ttl=255 time=0.854 ms
```

2025-11...15.53.49

Bmarco11 — D1 — telnet 192.168.71.128 5000 — 86×23

```
D1(config)#interface vlan 11
D1(config-if)#shutdown
D1(config-if)#exit
D1(config)#
*Nov  5 14:57:32.530: %HSRP-5-STATECHANGE: Vlan11 Grp 116 state Active -> Init
D1(config)#
*Nov  5 14:57:34.537: %LINK-5-CHANGED: Interface Vlan11, changed state to administrati
vely down
*Nov  5 14:57:35.543: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan11, changed
state to down
*Nov  5 14:57:35.544: %HSRP-5-STATECHANGE: Vlan11 Grp 11 state Active -> Init
D1(config)#interface vlan 11
D1(config-if)#no shutdown
D1(config-if)#exit
D1(config)#
*Nov  5 14:58:07.095: %LINK-3-UPDOWN: Interface Vlan11, changed state to up
*Nov  5 14:58:08.101: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan11, changed
state to up
D1(config)#
*Nov  5 14:58:09.254: %HSRP-5-STATECHANGE: Vlan11 Grp 11 state Listen -> Active
D1(config)#
*Nov  5 14:58:10.538: %HSRP-5-STATECHANGE: Vlan11 Grp 116 state Listen -> Active
D1(config)#
```

Bmarco11 — D1 — telnet 192.168.71.128 5000 — 86×23

```
*Nov  5 14:59:35.371: %HSRP-4-BADAUTH2: Bad authentication from 10.11.0.2
D1(config-if)#exit
D1(config)#interface vlan 21
D1(config-if)#standby 21 authentication md5 key-string Super53cret
D1(config-if)#exit
D1(config)#end
D1#
*Nov  5 15:00:04.381: %SYS-5-CONFIG_I: Configured from console by console
D1#
*Nov  5 15:00:05.551: %HSRP-4-BADAUTH2: Bad authentication from 10.11.0.2
D1#
*Nov  5 15:00:08.879: %HSRP-5-STATECHANGE: Vlan21 Grp 21 state Standby -> Active
D1#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State   Active          Standby         Virtual IP
Vl11        11   150 P Active  local           unknown         10.11.0.254
Vl11        116  150 P Active  local           FE80::D2:1      FE80::5:73FF:FEA0:74
Vl21        21   100 P Active  local           unknown         10.21.0.254
Vl21        216  100 P Standby FE80::D2:2      local           FE80::5:73FF:FEA0:D8
D1#
*Nov  5 15:00:35.682: %HSRP-4-BADAUTH2: Bad authentication from 10.11.0.2
D1#
```

Bmarco11 — D1 — telnet 192.168.71.128 5000 — 86×23

```
Interface   Grp  Pri P State   Active          Standby         Virtual IP
Vl11        11   150 P Active  local           unknown         10.11.0.254
Vl11        116  150 P Active  local           FE80::D2:1      FE80::5:73FF:FEA0:74
Vl21        21   100 P Active  local           unknown         10.21.0.254
Vl21        216  100 P Standby FE80::D2:2      local           FE80::5:73FF:FEA0:D8
D1#
*Nov  5 15:00:35.682: %HSRP-4-BADAUTH2: Bad authentication from 10.11.0.2
D1#
*Nov  5 15:01:05.863: %HSRP-4-BADAUTH2: Bad authentication from 10.11.0.2
D1#
*Nov  5 15:01:38.009: %HSRP-4-BADAUTH2: Bad authentication from 10.21.0.2
D1#
*Nov  5 15:01:44.634: %HSRP-5-STATECHANGE: Vlan21 Grp 21 state Active -> Speak
*Nov  5 15:01:45.427: %HSRP-5-STATECHANGE: Vlan21 Grp 21 state Speak -> Standby
D1#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State   Active          Standby         Virtual IP
Vl11        11   150 P Active  local           10.11.0.2       10.11.0.254
Vl11        116  150 P Active  local           FE80::D2:1      FE80::5:73FF:FEA0:74
Vl21        21   100 P Standby 10.21.0.2       local           10.21.0.254
Vl21        216  100 P Standby FE80::D2:2      local           FE80::5:73FF:FEA0:D8
D1#
```

Bmarco11 — D1 — telnet 192.168.71.128 5000 — 86×23

```
D1#conf
*Nov  5 15:03:41.003: %SYS-5-CONFIG_I: Configured from console by console
D1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
D1(config)#track 4 interface loopback 0 line-protocol
D1(config-track)#exit
D1(config)#interface vlan 11
D1(config-if)#standby 11 track 4 decrement 60
D1(config-if)#standby 116 track 4 decrement 60
D1(config-if)#exit
D1(config)#end
D1#
*Nov  5 15:05:42.137: %SYS-5-CONFIG_I: Configured from console by console
D1#show standby
Vlan11 - Group 11 (version 2)
  State is Active
    8 state changes, last state change 00:07:42
  Virtual IP address is 10.11.0.254
  Active virtual MAC address is 0000.0c9f.f00b (MAC In Use)
    Local virtual MAC address is 0000.0c9f.f00b (v2 default)
  Hello time 250 msec, hold time 750 msec
    Next hello sent in 0.064 secs
  Authentication MD5, key-string
```

Bmarco11 — D1 — telnet 192.168.71.128 5000 — 86×23

```
D1#show standby
Vlan11 - Group 11 (version 2)
  State is Standby
    10 state changes, last state change 00:01:24
  Virtual IP address is 10.11.0.254
  Active virtual MAC address is 0000.0c9f.f00b (MAC Not In Use)
    Local virtual MAC address is 0000.0c9f.f00b (v2 default)
  Hello time 250 msec, hold time 750 msec
    Next hello sent in 0.176 secs
  Authentication MD5, key-string
  Preemption enabled
  Active router is 10.11.0.2, priority 100 (expires in 0.784 sec)
    MAC address is aabb.cc80.0200
  Standby router is local
  Priority 90 (configured 150)
    Track object 4 state Down decrement 60
  Group name is "hsrp-Vl11-11" (default)
Vlan11 - Group 116 (version 2)
  State is Standby
    10 state changes, last state change 00:01:13
  Link-Local Virtual IPv6 address is FE80::5:73FF:FEA0:74 (conf auto EUI64)
  Active virtual MAC address is 0005.73a0.0074 (MAC Not In Use)
    Local virtual MAC address is 0005.73a0.0074 (v2 IPv6 default)
```